

Public Economics and Policy

Lecture 6

Economic Analysis of Taxation (Part I)

September 8, 2026

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Learning objective

- ☐ Understand the concept of taxes
- ☐ Analyze the impact of tax incidence in competitive markets and how demand elasticity affects tax incidence.
- ☐ Evaluate the tradeoffs between taxation efficiency and equity goals using Ramsey's insights.

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Outline

- ❑ Definition of taxes and forms of tax
- ❑ Tax incidence in competitive markets
- ❑ Effects of the demand elasticity on tax incidence
- ❑ Tradeoff between the efficiency and the equity goal in taxation – based on the Ramsey's basic insights

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Reading

- ❑ Stiglitz, J. E., & Rosengard, J. K. (2015). *Economics of the public sector: Fourth international student edition*. WW Norton & Company.
(Chapter 17)

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What is tax? Why do we pay tax?

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Definition of a Tax

- ❑ Tax is a kind of mandatory recurring fee which is paid to the central and state/local government.
- ❑ Tax is regarded as the main source of revenue for the government (Govt does not sell products or have profits, so the only way to fund services is by asking us to pay taxes on the money we earn, things we buy, and property we own.)
- ❑ An individual or any corporate entity are obliged to pay taxes.
- ❑ Taxation reallocates resources from private to government use in two steps.
 1. *The ability of individuals to command resources is reduced*, because taxation reduces income for spending on market goods and services.
 2. *The revenues collected by government are used to provide government goods and services* and to provide income support payments to recipients of government transfers such as Social Security pensions.

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What is the difference between a tax avoidance and a tax evasion?

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Tax avoidance vs. Tax evasion

Tax avoidance

☐ Legal by exploiting loopholes in tax laws and is considered morally questionable.

Tax evasion

☐ Illegal as it involves fraudulent practices to reduce tax burden.

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Example

☐ Tax deductions, life insurance premium	
☐ Underreporting the sales revenue	
☐ Claiming dependents or credits that one is not entitled to	
☐ Establishing an offshore company in a tax haven	

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Tax Havens

- ☐ Tax havens are places with special tax rules that make them attractive to individuals and companies. They allow people to pay less in taxes, protect their money, and bring in investments from around the world.
- ☐ Tax havens can offer rebates for taxes or tax incentives for attracting outside investment.
- ☐ The other attributes for tax havens include the protections of personal finance information and they do not require outside entities to have a substantial local presence
- ☐ Example countries that also rank high in secrecy and have low-to-no taxes are the British Virgin Islands, the Cayman Islands and Bermuda.

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Tax Havens

- ❑ Tax havens typically do not require outside entities to have a substantial local presence --- For example, a 2008 Government Accountability Office report found that one building in the Cayman Islands housed 18,857 mostly international companies.
- ❑ One can claim tax benefits by merely hanging a nameplate in a tax haven & no need for actually producing goods or services or conducting trade or commerce within the boundaries of the country
→ Tax evasion
- ❑ The Cayman Islands have some of the best secrecy laws, while other countries that also rank high in secrecy and have low-to-no taxes are the British Virgin Islands, Bermuda, Belize, Switzerland, and the United Arab Emirates.

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Tax Havens

Adverse impacts of tax havens

- ❑ Revenue Loss for Governments
 - ❑ For example, in 2016, the European Commission ordered Apple to repay €13 billion in unpaid taxes to Ireland, as the company had taken advantage of Ireland's low corporate tax rate. Apple appealed the decision and won their argument in 2020, relieving them of owing taxes.
- ❑ Inequity and Inequality
 - ❑ Tax havens can exacerbate income inequality. Wealthy individuals and large corporations often have the resources to research and take advantage of these tax breaks.
 - ❑ For instance, in the Panama Papers leak in 2016, it was revealed that wealthy individuals and companies from around the world used Panamanian law firm Mossack Fonseca to set up offshore accounts to avoid taxes.
 - ❑ This highlights how the global elite were using tax havens, many of which would not otherwise be accessible by normal citizens.

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Forms of Taxation

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Forms of Taxation

- ❑ Taxes can be divided into two broad categories: Direct taxes and indirect taxes

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Forms of Taxation: Direct Tax

- ❑ A tax where the taxpayer (individuals and businesses) **pays directly to** the authority imposing the tax.
- ❑ **Key:** The taxpayer has to bear the tax (levied on income and profits) and will not be able to transfer this liability to another entity.
- ❑ Direct tax includes: personal/incorporate income tax, capital gains tax, inheritance tax, land and building tax
- ❑ Tax rate depends on income and profit.
- ❑ Individual income tax (levied both on wages and on capital income) affects decisions about labor supply and decisions about savings.
- ❑ The incidence of **tax evasion** is possible.

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Note

- ❑ Poor level of voluntary compliance with tax laws by taxpayers and predominance of cash transactions are likely to be associated with tax evasion.
- ❑ It would be unlikely to track down on the value of economic activities if the payments are based on cash /cryptocurrencies.
- ❑ The advantage of E-payment economy is that the value of economic activities is traceable.
 - ❑ In Thailand, individuals' money transfer (Only money transfer into the account) of all accounts in 1 year (from 1 January to 31 December of that year) will be reported to the tax office if:
 - ❑ Frequent transactions (at least 3,000 times without specifying the total amount of money transfer transactions into the account)
 - ❑ At least 400 times of money transferred into the account and the total value is at least 2 millions baht.

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Forms of Taxation: Indirect Tax

- ❑ Indirect Tax are charged on goods and services.
- ❑ The taxpayers pay the indirect tax to the government via intermediary.
- ❑ Indirect Tax includes: Goods and Services Tax (GST)/value added tax, excise duty, customs duty
- ❑ Tax rate is the same for everyone. As a result, tax burden is regressive.

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Direct Tax vs. Indirect Tax

- ❑ Indirect taxes tend to increase the tax burden on the lower income earner. (An individual cannot not pay them even if he has little income.)
 - ❑ For a purchase of the same commodity, a high-income earner and a low-income earner pay the same amount of GST/VAT.
- ❑ For the PIT, the high and the low-income earners pay different amount of tax at different rate of tax.
 - ❑ With respect to the ability to pay taxation, the PIT is such that tax revenue is be collected from those who have the ability to pay it. → indicating **equity in the tax system**
 - ❑ In the case of a direct tax dominated tax system, there is equity in the sense that the tax contribution comes from those who have the ability to pay (i.e., the rich).

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Tax Base

- ❑ The tax base is the total amount of income, property, assets, consumption, transactions, or other economic activity subject to taxation by a tax authority
 - ❑ If the government could ascertain who had greater abilities, and who therefore was in a better position to pay taxes → impose higher taxes on those individuals.
 - ❑ But, how can abilities be measured? → not observable
 - ❑ So, a tax depends on easily measured variables such as expenditures or income.
- ❑ Three broad categories of tax base: income, consumption, and wealth
- ❑ Example of tax base of PIT in Thailand:
 - ❑ population of almost 70 million people
 - ❑ Out of 9.55 million taxpayers, there were 3.3 million people who had a monthly income of 25,000 baht (the minimum threshold to file tax forms)

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Indirect tax

- ❑ Indirect taxes (VAT/GST/Sale tax, excise tax, import duty) are usually collected by intermediaries, such as producers or retailers.
- ❑ Producers can pass on the tax burden to the final consumers.
- ❑ Indirect taxes can affect the prices and quantities of goods or services in the market, and have different effects on consumers and producers. → ***Burden of tax on consumers vs. Burden of tax on producers***
- ❑ Two types of tax: Ad valorem tax vs. Specific tax

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Ad valorem tax

- ❑ A tax based on the assessed value of an item (proportional to the price)
- ❑ Ad valorem tax is more progressive than a specific tax, as it places a higher tax burden on more expensive goods.
- ❑ An ad valorem tax may induce consumers to buy less expensive goods to avoid paying higher taxes.

Specific tax (Per unit tax)

- ❑ A fixed amount of tax per unit of a good or service, regardless of its price
- ❑ A specific tax increases the price of all varieties equally, and has a bigger effect on reducing the overall demand

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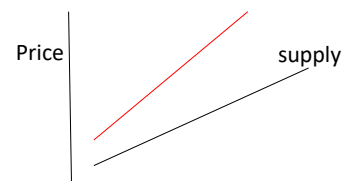
Ad valorem tax vs. Specific tax

- ❑ **Ad valorem tax** is levied as a given percentage of the price (e.g. the excise tax on sugary beverages = 10% of the price excluding VAT)

$$P = 12, \text{ After-tax price} = 12 + (0.1 \times 12) = 12 + 1.2 = 13.2$$

$$P = 15, \text{ After-tax price} = 15 + (0.1 \times 15) = 15 + 1.5 = 16.5$$

$$P = 20, \text{ After-tax price} = 20 + (0.1 \times 20) = 20 + 2.0 = 22$$

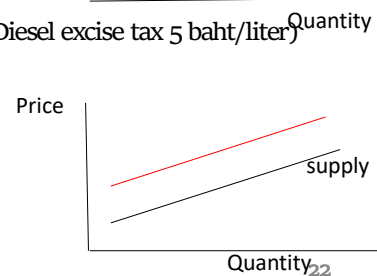


- ❑ **A specific tax (Per Unit tax)** is a fixed amount per unit output (e.g. Diesel excise tax 5 baht/liter)

$$P = 12, \text{ After-tax price} = 12 + 5 = 17$$

$$P = 15, \text{ After-tax price} = 15 + 5 = 20$$

$$P = 20, \text{ After-tax price} = 20 + 5 = 25$$



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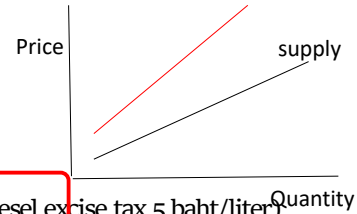
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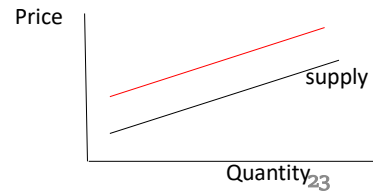
For the simplicity in an analysis, we will apply a specific tax.

❑ **A specific tax (Per Unit tax)** is a fixed amount per unit output (e.g. Diesel excise tax 5 baht/liter)

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$$P = 20, \text{ After-tax price} = 20 + 5 = 25$$



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Share your thoughts on ...

❑ Suppose that a govt imposes an excise tax on sugary beverages and the tax is legally imposed on the sellers.

❑ Question:

(1) Would the seller bear all the burden of a tax?

(2) Or, would the burden of a tax be shared between consumers and sellers.

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Incidence of a tax

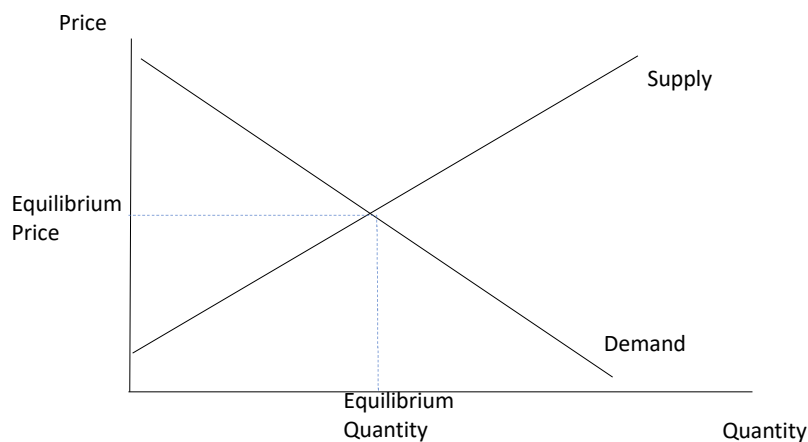
- ❑ The effect of the tax on the prices consumers pay and sellers receive in a market.
- ❑ The burden of a tax --- the difference between the price paid/received before and after the tax has been imposed.
- ❑ The burden of the tax is shared by both consumers and producers
 - ❑ Equally shared between consumers and producers? *Not likely*
 - ❑ Size of tax burden on consumers and producers depends on the elasticity of demand and the elasticity of supply.

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Equilibrium in the market

Equilibrium condition: Demand = Supply

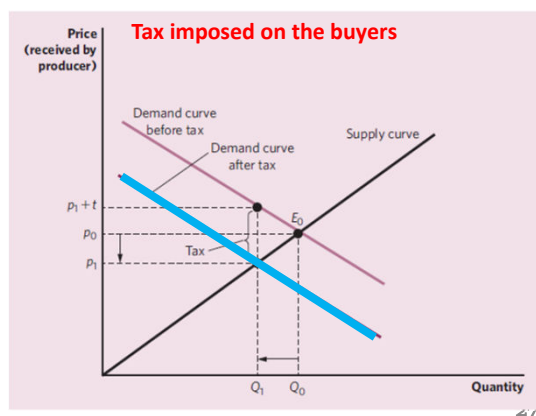
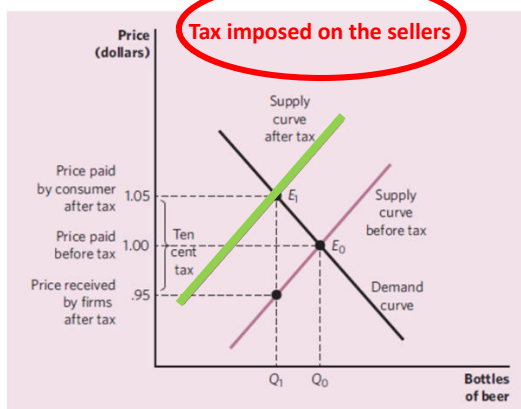


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Tax incidence in competitive markets

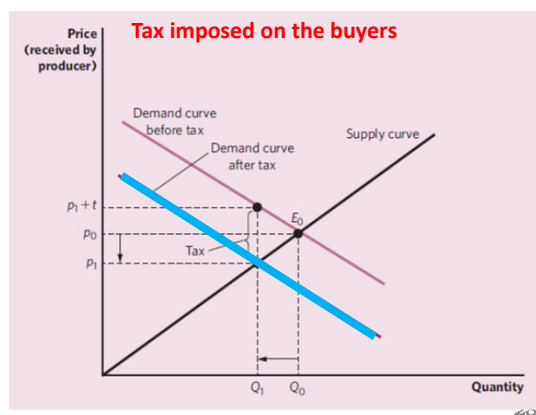
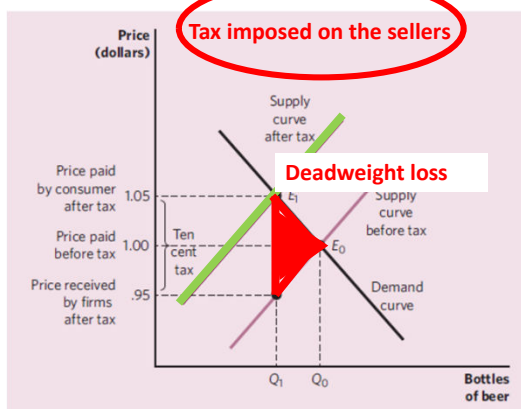
- No difference whether a tax on a commodity is imposed on the commodity's producers or on its consumers (Figures 18.2 and 18.3)



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Tax incidence in competitive markets

- No difference whether a tax on a commodity is imposed on the commodity's producers or on its consumers (Figures 18.2 and 18.3)



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Deadweight loss

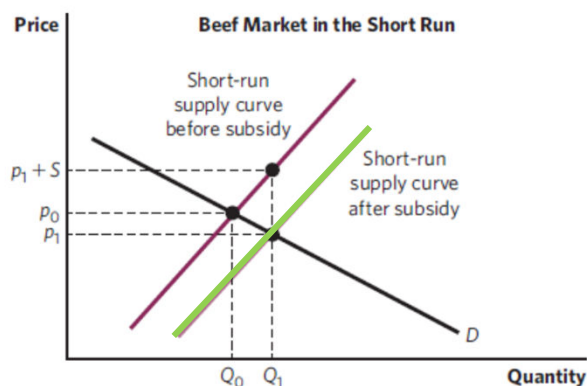
- ❑ **A deadweight loss** is value that simply vanishes — nobody ends up holding it. That's what makes it different from money changing hands. If the government takes 12 baht from you in tax, you're poorer and the government is richer: the money moved. But if something stops a deal that *both* sides wanted, the benefit that deal would have created doesn't move anywhere. It just never comes into existence.
- ❑ Think of it as the **"trades that didn't happen"** pile. Every time a buyer who values something at more than it costs to make walks away, a small gain that was sitting there for the taking disappears — and no one gets it: not the buyer, not the seller, not the government.
- ❑ **Example — the coffee you didn't buy.** A café can make a cup for 40 baht and would happily sell at 45. You'd happily pay 52. If you buy at 45, you gain 7 baht of value and the café gains 5 — 12 baht of value created out of a two-minute transaction. Now a 20-baht tax is added, so the price must be at least 60. You don't buy. You go without coffee, the café loses the sale and the barista's hour of work, and the government collects nothing, because nothing was sold. That 12 baht of value didn't move — it evaporated. That's the deadweight loss, and it's exactly what the red triangle in the Figure is adding up.

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The incidence of a government subsidy

Refer to the Figure on page 554,



- ❑ Effects on the price
- ❑ Effects on the quantity

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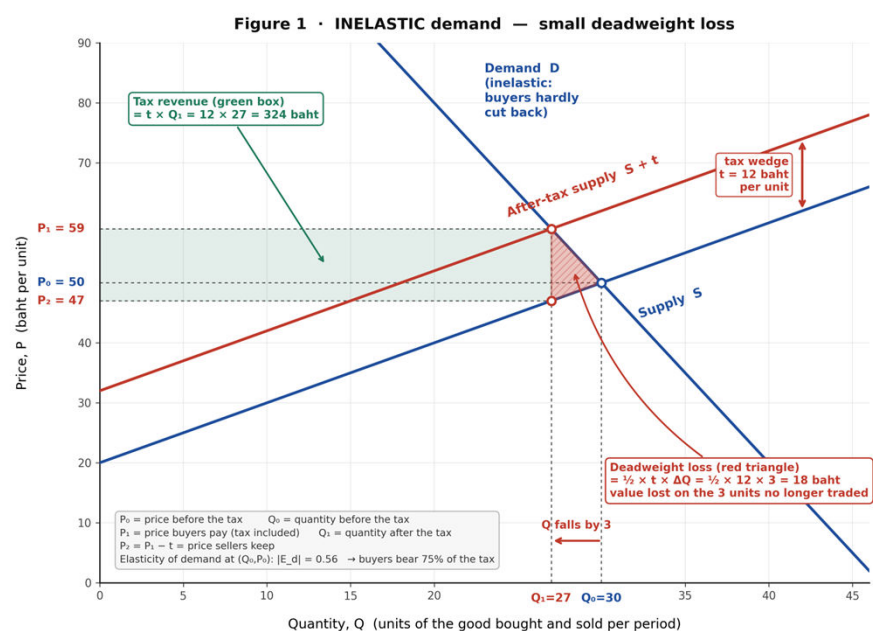
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Tax distortions and burden

□ The elasticity of demand determines the size of tax distortions.

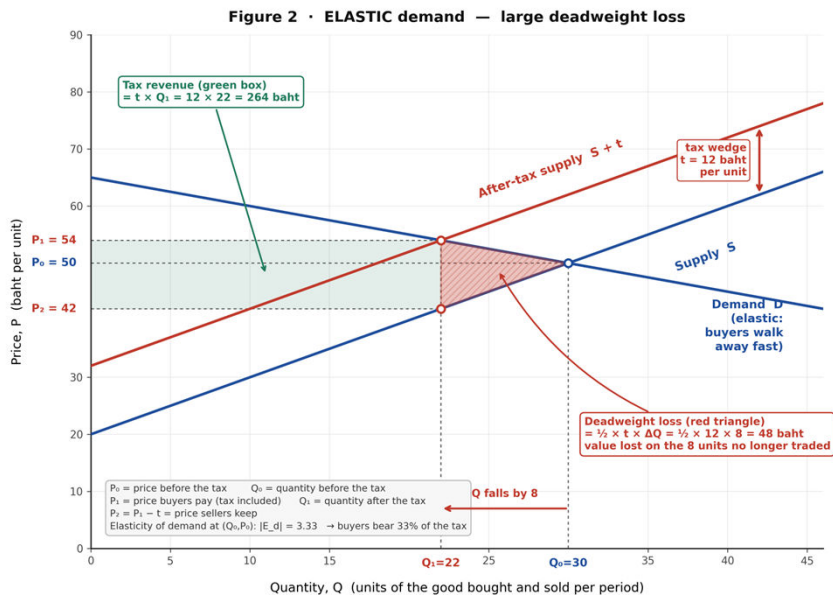
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□ Recall that

- The demand for necessity goods is **inelastic**. $\rightarrow$ A small change in price leads to a small change in quantity demanded & consumers are not very sensitive to the price changes.
- The demand for luxury goods is **elastic**. $\rightarrow$ A small change in price leads to a large change in quantity demanded & consumers are very sensitive to the price changes.
- When imposing a tax on the commodity with inelastic demand (such as food), the tax distortion is smaller.
- To achieve the efficiency in taxation, would you recommend a govt to impose a higher tax rate on food and a lower tax rate on wines?

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Two sides of the same coin

- ❑ By doing so, the govt would achieve the efficiency in taxation.
- ❑ But, what about the fairness?

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Ramsey Taxes

- ❑ Ramsey's basic insight:
 - ❑ **The condition for optimal commodity tax:** Commodities with *low elasticity of demand have a lower marginal deadweight loss per marginal dollar of revenue raised*, and thus should face higher marginal tax rates.
 - ❑ **High tax rates should be imposed on commodities with low price elasticities (like food) & Low tax rates should be imposed on commodities with high price elasticities (like luxuries) → Efficiency is achieved.**

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Ramsey Taxes

□ Ramsey's basic insight:

□ High tax rates should be imposed on commodities with low price elasticities (like food) & Low tax rates should be imposed on commodities with high price elasticities (like luxuries)

□ *If a high tax is imposed on commodities with low price elasticities (like food), then the poor will bear a larger burden than the rich.*

→ Fail to achieve the equity goal

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Ramsey Taxes

□ Subsequent research has extended Ramsey's original analysis to include redistributive goals. → e.g. taxing income-elastic and price-elastic commodities, such as cosmetics at a higher rate than income-inelastic and price-inelastic commodities like food if concerning for *income redistribution*.

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Conclusion

- ❑ Tax is regarded as the main source of revenue for the government & An individual or any corporate entity are obliged to pay taxes.
- ❑ Taxes can be divided into two broad categories: Direct taxes and indirect taxes.
- ❑ Tax burden on buyers and sellers can be different depending on the demand elasticities.
- ❑ From the Ramsey's basic insight:
 - ❑ The condition for optimal commodity tax: Commodities with low elasticity of demand have a lower marginal deadweight loss per marginal dollar of revenue raised, and thus should face higher marginal tax rates.
 - ❑ High tax rates should be imposed on commodities with low price elasticities (like food) & Low tax rates should be imposed on commodities with high price elasticities (like luxuries) Efficiency is achieved.

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